

Md. Ismail Hossain

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Biostatistician with dual MS degrees in Applied Statistics and Applied Mathematics and 5+ years of experience in addiction research, cardiovascular outcomes, metabolomics, and clinical biomedical studies. Co-authored 8+ peer-reviewed publications (2024–2025), including a study in *Nature Mental Health* as lead Biostatistician. Proficient in R and Python for advanced statistical modeling, machine learning, and AI applied to complex biomedical datasets.

EDUCATION

New Mexico Institute of Mining and Technology <i>MS Mathematics (Operations Research and Statistics)</i>	Socorro, New Mexico July 2023
University of Dhaka <i>MS Applied Statistics</i>	Dhaka, Bangladesh Dec 2015
University of Dhaka <i>BS Applied Statistics</i>	Dhaka, Bangladesh Dec 2013

FULL TIME EXPERIENCE

Data Analyst (Institutional Research & Biostatistics) <i>University of Wyoming</i> - Built Power BI and R dashboards to track institutional metrics, research proposals, awards, and funding, enhancing transparency for university leadership. - Applied Prophet time series modeling to indirect cost data to support financial forecasting and strategic resource allocation. - Benchmarked UWyo's national R&D standing using regression analysis of faculty counts and expenditures across peer institutions, informing strategic investment priorities.	Dec 2024 – Nov 2025 Laramie, Wyoming
Biostatistician <i>Louisiana State University Health Shreveport</i> - Led statistical analysis for a study published in Nature Mental Health; served as lead biostatistician across 8+ peer-reviewed publications (2024–2025). - Designed a cost-effective bioinformatics (metabolomics) pipeline, reducing per-dataset analysis costs by \$20,000. - Taught Advanced Biostatistics (IDSP 227) across two semesters and mentored 5 graduate and medical students on independent research projects.	Sep 2023 – Nov 2024 Shreveport, Louisiana
Graduate Research & Teaching Assistant <i>New Mexico Institute of Mining and Technology</i> - Assisted faculty in teaching undergraduate and graduate statistics and mathematics courses; supported course design and student assessments. - Conducted independent research in operations research and statistical modeling as part of MS Mathematics program.	Aug 2021 – July 2023 Socorro, New Mexico
Senior Data Scientist <i>Insightin Technology Bangladesh Ltd., Data Science Team</i> - Engineered an automated Python data quality pipeline processing ~20GB of data, generating accurate PDF reports and cutting runtime from 3 hours to 15 minutes – a 12× speedup. - Applied NLP-based sentiment analysis to large-scale customer feedback data, identifying key dissatisfaction drivers and contributing to a 5% increase in annual revenue.	Dec 2018 – Jul 2021 Dhaka, Bangladesh
Early Career Data Roles <i>Innovision Consulting & Therap BD</i> Performed data management, SQL analysis, Power BI dashboard QA, and real-time survey monitoring across M&E and BI teams.	2016 – 2018 Dhaka, Bangladesh

FREELANCE CONSULTING

Statistical Consultant (Upwork / PeoplePerHour) <i>Statistical consulting, biostatistical analysis, data modeling, and BI dashboards for 193 international clients</i>	July 2013 – July 2021 Remote
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SELECTED PROJECTS

Medical Image Denoising Using Total Variation and Convex Optimization (2023) - Implemented and compared ADMM and Fast Gradient Projection (FGP) to denoise brain MRI images affected by Gaussian noise using MATLAB. - Evaluated denoising quality via SSIM; ADMM outperformed FGP at low noise levels (0.05), with comparable performance at higher levels (0.1).

RESEARCH INTERESTS, DATABASES & SKILLS

Expertise & Methods: Biostatistics & Clinical Data Analysis, Machine Learning & AI in Healthcare, Longitudinal & Survival Analysis, NLP & Unstructured Data Modeling, Clinical Research & Study Design; Regression, Bayesian Methods, Propensity Score Matching, TMLE, Deep Learning.

Domains, Data & Tools: Multiomics, Clinical Trials, Transcriptomics, Addiction, Cardiovascular, Brain Research; NHANES, NIS, HCUP, All of Us, EMR/Epic; R, Python, SQL, SAS, Stata, SPSS, Tableau, Power BI, AzureML, PySpark, AWS.

1. **Boullion, Jolie; Hossain, Md Ismail, et al.** “Machine Learning–Based Biomarker Identification for Early Diagnosis of Metabolic Dysfunction–Associated Steatotic Liver Disease,” *Journal of Clinical Endocrinology & Metabolism* (2025).
2. **Mubashir, Maryam; Hossain, Md Ismail, et al.** “Impact of Interstitial Cells of Cajal Density and Other Factors on Clinical Outcomes Following G-POEM in Gastroparesis Patients,” *Digestive Diseases and Sciences* (2025).
3. **Kelley, Roger E.; Hossain, Md Ismail, et al.** “Diagnostic Yield and Therapeutic Implications of Vascular Imaging in Acute Ischemic Stroke,” *Journal of Stroke and Cerebrovascular Diseases* (2025).
4. **Thapa, Monika; Hossain, Md Ismail, et al.** “The Correlation of Whole Blood Viscosity and Outcome in Mechanical Thrombectomy for Acute Ischemic Stroke,” *Frontiers in Stroke* (2025).
5. **Xing, Diensn G; Hossain, Md Ismail, et al.** “Prevalence and Patterns of Methamphetamine Use and Mental Health Disparity in the United States,” *Nature Mental Health* (2024).
6. **Raza, Syed; Hossain, Md Ismail, et al.** “GERD in Gastroparesis Patients after Gastric Per Oral Endoscopic Pyloromyotomy – A Long-Term Follow-Up,” *Gastrointestinal Endoscopy* (2024).
7. **Ali, Shafaqat; Hossain, Md Ismail, et al.** “Trends in Peripheral Artery Disease and Critical Limb Ischemia Hospitalizations Among Cocaine and Methamphetamine Users,” *Frontiers in Cardiovascular Medicine* (2024).
8. **Husein, Amanda; Hossain, Md Ismail, et al.** “Trends and Patterns in Pulmonary Arterial Hypertension-Associated Hospital Admissions among Methamphetamine Users,” *Frontiers in Cardiovascular Medicine* (2024).